Chapter 6

Knowledge Based Expert System

In this chapter we will look at the Expert system which can guide and assist an individual once it is provided with the enough Knowledge to do so. Here we will discuss about expert system and its architecture with proper steps of making an expert system in any field by several tools which are available to do so. For expert systems the most important is Knowledge Engineering which is the collection of Knowledge from an expert and Engineering it in such a form that a machine could understand it.

Introduction

One of the pioneering fields of AI study is Expert Systems (ES). It is proposed by researchers in the computer science department of Stanford University. An expert system is a computer system which emulates a human expert's decision-making skills. Expert systems are developed by reasoning through information structures, representing laws, rather than through traditional procedural code, in order to overcome complex problems.

Expert Programs use professional expertise systematically to address human expert problems. An expert is an individual with experience in a specific field. The expert has expertise or special credentials which most people do not learn or care about. An expert can solve or solve problems more effectively than people can solve. Expert System actually emulates all these qualities of expert and solve the problem in a better way and fast.

Today, Expert System (ES) is used in the fields of finance, research, engineering, technology, medicine and many others where there is a well-defined problem area. The fundamental principle is that an expert method should also define the steps to illustrate why a problem can be solved. Unless an individual can justify his reasoning for Expert System.

Expert System always works on basis of its problem. In expert System designing we always define the problem domain and latter on we will collect knowledge related to it. As you can see in the below figure knowledge domain of an Expert system is always a subset of problem domain as represented in the Fig 6.1. The knowledge which is being built is always gathered from books, magazines, papers or experts based on their experience in particular field.

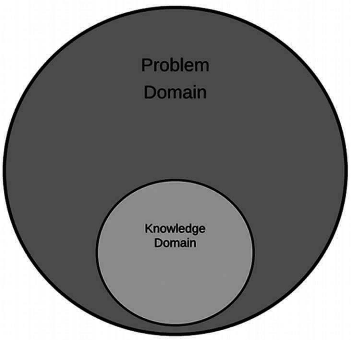

Fig. 6.1 Expert System Knowledge domain

All the knowledge which is gathered will be based on the problem defined and it will specific and based on facts.

Characteristics of Expert System

Knowledge is the most critical aspect of any expert program. The power of the expert system lies in the high-quality awareness of task areas. In expert systems, knowledge is separated from its processing i.e., the knowledge base and the inference engine are split up. For storing this information, a standard system is a mixture of information and the control structure. The combination creates issues with the system code's comprehension and analysis, as any alteration in the code affects information and processing. The expert method includes a knowledge base and a set of rules to apply the knowledge base to each unique situation defined in the program. Comprehensive expert systems may be improved by adding to the knowledge base or the collection of rules. Expert system can be developed from scratch or designed with a software development part called a "kit" or shell. A shell is a full development environment for the creation and maintenance of application based on knowledge. It offers a step-by-step approach and an preferably user-friendly interface for a knowledge engineer who can directly involve domain experts in structuring and coding the information

Expert system as seen can be developed and they have some general characteristics which are

- **High Performance:** The system must be capable of responding at very high level of competency or better than that of any expert in the field. i.e.,, the advice quality given by the system should be high and better as compart to the expert.
- `Adequate Response time:` The Expert should perform or response to a situation quicker as compared to expert and with an accurate and better advice or solution to the query.
- **Good Reliability:** The expert system should not be prone to lags and crashes it should be well functioning in any situation.
- **Understandable:** The Expert system should able to explain its way of decision making on query so that it can be understandable.

- **Flexibility:** The Expert System contains large amount of data in the form of knowledge regarding or related to certain field and it should be very easy to add, change and delete any knowledge into the system for its improvement.

Architecture of an Expert System

An expert system tool, or shell, includes the essential components of expert systems. The knowledge base and the Inference engine are the main components of the expert systems. and other Elements of Expert System which can be seen in the Fig 6.2 are:

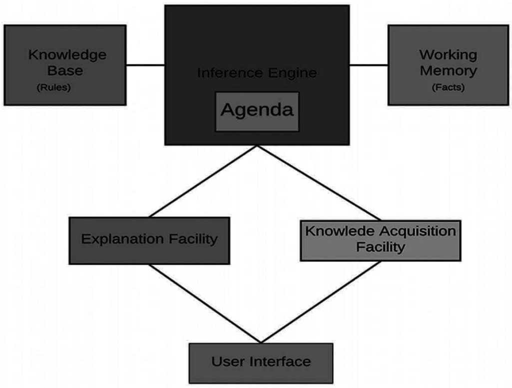

Fig. 6.2: Architecture of Expert System

- **Knowledge Base:** The knowledge base includes the facts to be learned, formulated and solved. It is a domain information store captured by the human expert through the module for learning. It uses rules, frames, logic, semantic network, etc to reflect the information output. The expert system knowledge base provides empirical and heuristic knowledge. True knowledge is the general knowledge of the task domain, commonly contained in textbooks or journals. Heuristic information is a less rigorous, experiential, judgmental, and generally individualistic awareness of success. It includes knowledge of good practice, good judgment, and logic possible in the area.
- **Inference Engine:** It is an expert System's brain. It uses the control system and offers a reasoning technique. This serves as an interpreter who analyzes and operates on the rules. This is used to suit histories of the responses and firing rules received by users. The key task of the inference engine is to map the path to a conclusion through a set of rules. Two methods, i.e., forward and rear chaining, are used here.
- Working Memory
- Agenda: It contains the list of the task or queries which need to be performed by expert system.
- **Explanation facility:** It is a subsystem which explains the behavior of the system. The description can differ from how to arrive at the final or intermediate solutions to explain the need for further detail. Here users want to ask simple questions as to why and how the device information is being shared with the user and acts as a tutor.
- **Knowledge Acquisition Facility:** The acquirement of knowledge is a buildup, transition and transformation into a computer system to create or expand the

knowledge foundations of problem-solving expertise from experts and/or recorded information sources. It is an expert subsystem for developing information bases. For the acquisition of information, process analysis, interviews and observation techniques are used

- **User Interface:** It is an interaction device with the user. It provides facilities for user-friendly conversation, such as menus, graphical gui etc. The User Interface is responsible for translating rules from its internal (which users do not understand) representation into user comprehensible types.

Knowledge engineering is known to develop the expert System. Domain experts, users, knowledge engineers and system maintenance staff are employees involved in the expert system development. Field specialists have unique expertise, assessment, practice and strategies for offering guidance and solving problems. It provides information on job success. During the creation of the inference engine, the knowledge base framework and the user interface, the knowledge engineer is involved. The Knowledge Engineer and the specialist in information will foresee the needs of the consumer in creating a program

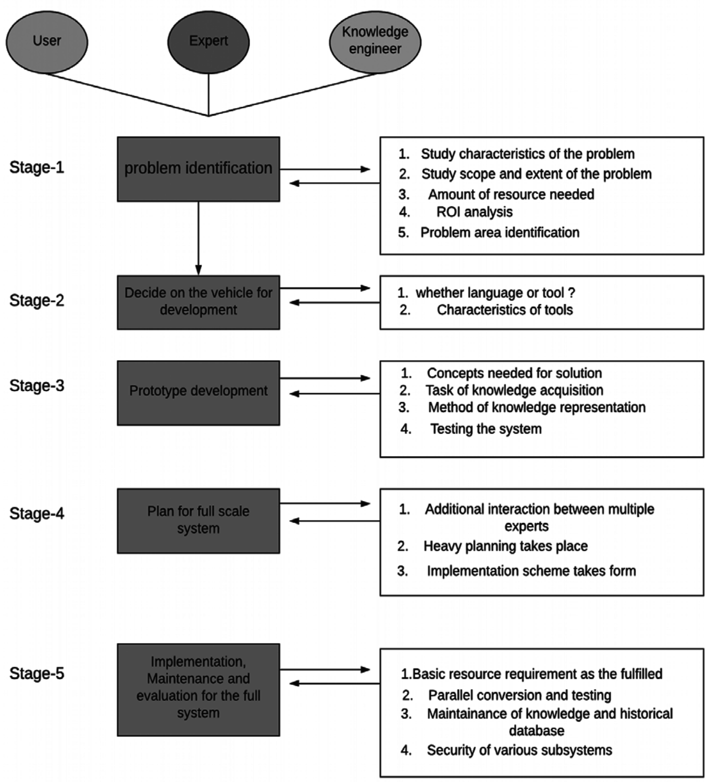

Fig. 6.3 Expert System Life Cycle

The development of an expert system includes five key phases as seen in the Fig 6.3. Each stage has its own unique characteristics and is associated with other stages.

Stage 1: Problem Identification: In this step, the expert and the Knowledge Engineer communicate to detect the problem. The key issues previously discussed are investigated for the problem's characteristics. The degree and nature are examined. The quantity of resources available, such as people, computer resources, finances etc. The ROI analysis is conducted. Problem areas that can cause a great deal of trouble are defined and the overall design and conceptual solution to that problem is established.

Stage 2: Deciding Mode of Development: The immediate phase, when the problem is found, is to determine which model will be built. The software technician may use a programming language like PROLOG and LISP or some modern language to develop the framework or use a development shell from scratch. During this point, different shells and instruments for their suitability are described and analyzed. Tools whose properties suit the problem are thoroughly evaluated.

Stage 3: Prototype Development: The following are the required activities before creating a prototype-

- Decide on which concepts the solution would have to produce. The level of information (granularity) is an essential factor to be calculated. The development of the system starts with coarse granularity and progresses to fine granularity.
- Then starts the process of gaining information. The knowledge engineer and the field expert often communicate, and domain information is extracted.
- The knowledge engineer determines the form for representation until the expertise has been acquired. A conceptual image of the representation of information should have emerged during the identification process. This view is implemented or updated in this process.
- A prototype is developed when the scheme of information representation and information is available. Each prototype is evaluated for different problems and the prototype is reviewed. This method results in knowledge of fine granularity, which is effectively encrypted in the basis of information.

Stage 4: Full System planning: The success of the prototype gives the full-scale device impetus. The region of the problem that can be fairly easily applied is first selected in prototype design. The creation of a subsystem is allocated to group leaders during full implementation, and schedules are drawn up. Gantt map, PERT or CPM are to be used.

Stage 5: Implementation, Evaluation and Maintenance: It is an expert system's final life cycle process. At the site, the entire scale system is created. The basic criteria for services are met at the site and simultaneous conversion and testing strategies are adopted. The final device is rigorously checked and ultimately passed to the customer.

Evaluation in any AI system is a challenging task. Solutions to AI problems are, as already described, only satisfactory. Since the evaluation requirements are not available, evaluation is difficult. What can be done is to provide the program and human expert with a series of problems and to compare the results.

System maintenance requires changing the knowledge base since there is no established information, environment, or issues. It is important to maintain the historical records and keep track of minor modifications made to the Inference engine. Maintenance also includes security.

Tools for Development of Expert system

There are some tools which are available for development of an expert system as represented by fig 6.4. These tools are divided into four categories which are:

(a) Programming Languages

(b) Knowledge Engineering Languages

(c) System Building Aid

(d) Support facilities

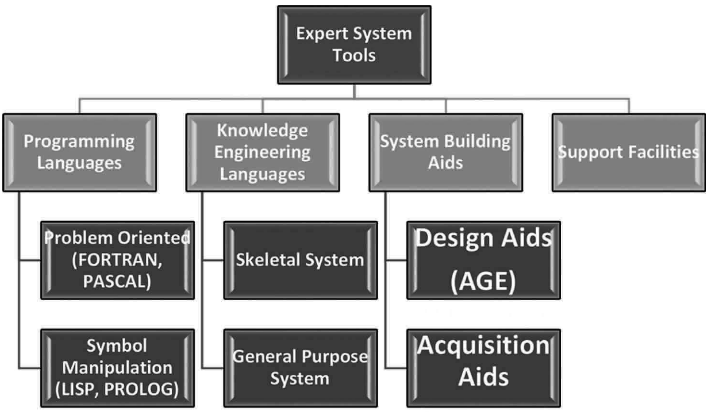

Fig. 6.4 Tools of Expert system

(a) **Programming Languages:** There are certain languages which are available and can be used to develop an expert system. It always based on the type of the problem we have like

i. **Problem-Oriented languages:** Such as Fortran and pascal. These are designed for a particular class of problems. Fortran have a convenient feature for performing algebraic calculations.

ii. **Symbolic Manipulation Languages**: such as LISP and PROLOG. LISP has the mechanism which can manipulates the symbol available in the form of list structures.

(b) **Knowledge Engineering Languages:** A language of knowledge engineering is a sophisticated method to build systems for experts consisting of an expert system that builds language in an extensive support environment.

i. **Skeleton System:** A skeletal language of information creation is a simplified expert system. Domains are removed from the expert program, leaving only the Inference engine and support facilities. For example, MYCIN is tripped down to EMYCIN which is easy and fast.

ii. General Purpose System: A Knowledge Engineering language can handle many problem areas and forms for a general purpose. This offers more power than skeleton framework over data access and search, but is much harder to use.

It varies according to the flexibility and generality. The Skeletal and General-purpose system falls in the research system category.

(c) **System Building Aids:** Consists of programs that help to develop the expertise of the industry expert and programs that help design the Expert System construction method.

i. Designing Aid: It helps in designing the expert system.

For example: Like age

It Assist the design and development of an expert System by the knowledge engineer.

HEARSAY, first used in the mid-1970s, interprets the enemy data on radio contact HEARSAY-III HANNIBAL. This uses data position information and signal characteristics to classify organizational units and their order of battle.

Provides the consumer with a collection of components that can be assembled as building blocks to form parts of an expert structure.

An expert system structure like forward chains, backward chaining, or blackboards architecture supports each component, a compilation of INTERLISP functions.

ii. Knowledge Acquisition Aid: It helps in Knowledge acquisition for the Expert system.

For example: TEIRSIAS

Its system design helps transfer information to a knowledge base from a domain expert.

Only used for management of the database.

The program learns new rules for the problem area through an interaction allowing users to define rules in a specific subset of English.

This analyzes the rules, determines how accurate and clear these rules are and lets the user fix them.

(d) **Support Facilities:** Consist of tools for helping programming – like debugging aids, Knowledge base editors to enhance capabilities of a finished system and Extra software packages.

The main components of Support Facilities are

i. **Debugging aids:** It is divided into different types like

Trace facilities: shows name or rule no of fired rules list, sub-routines called.

Break packages: allow the user to tell where to stop program execution so that the user can stop it just before an error occurs.

Automated testing: allows the user to test a program automatically on several benchmarks

i. I/O facilities: It is divided into different types like

Run time Knowledge Acquisition: the tool itself enables the user to converse with the system and to seek the information required in the knowledge base.

Menus: provide the user with the menus to choose for inputs. Menus: No specific code required.

Accessibility of the operating system: enables ES to track and manage other work during service

(e) **Explanation facilities:** It is divided into different types like

Retrospective: it explains how the system reached a particular state. How system arrived at a conclusion

Hypothetical: explain what would have happened if certain fact or rule had been different.

Counterfactual: why an expected conclusion has not reached

(d) **Knowledge base editors:** It is divided into different types like

Automatic book keeping: It helps to monitor the changes made by the user.

Syntax checking: It checks whether the rules are entered in correct format and are in accordance with the grammatical structure of ES languages.

Consistency checking: It checks the semantics of rule entered to check if there is any conflict with the existing knowledge base.

Knowledge Extraction: This helps to add new rules or modify the rules

Pros and Cons of Expert System (ES)

It has a lot of attractive feature.

- **Increased Availability:** It increases the availabilities of its expertise in mass on any suitable computer hardware.
- Re**duced Cost:** The cost of expertise providing every user is decreased a lot.
- Reduced Danger: It can be utilized in any hazardous or remote location.
- **Permanence:** Once the expert system is developed it is permanent whether expert of that domain can be lost but expert system will last ever.
- **Multiple Expertise:** The Knowledge of several Expert system can be made work simultaneously and regularly on a given problem until it gets solved at any time of night or day. The level of expertise of an expert system is much higher than the human expert of that domain.
- **Increased Reliability:** Expert system are steady, unemotional and always complete the response at time.
- **Explanation:** The Expert System provide all the description how it reached to the conclusion on a query or problem. In the same case human expert cannot provide all the time explanation due to its tiredness or ill conditions.

- **Fast Response:** Expert system is capable fast and real-time responses all the time.
- **Intelligent tutor:** Expert system can become a tutor for student to study apply basic programs and explains its outcome.
- **Intelligent database:** It can be used to access databases intelligently.

Cons of Expert System

- Expert system has shallow Knowledge.
- It has no deep understanding of the concepts and their relationship until we provide.
- It doesn't have common-sense
- It is a closed world with specific knowledge.
- Expert system may not have or select the most appropriate method for solving a particular problem.
- Some Easy problem in expert system are computationally becomes expensive

Summary

Expert System has become a very pioneering branch of Artificial intelligence with several uses in several fields. After studying about expert system, we can see what is an expert system, how expert systems are built and what are the tools for making an expert system. Expert system in agriculture field can help the farmers in lot many ways by guiding them with proper knowledge to use advance technology and other herbal to increase their crop production and protect it from pests.

PART B: IMPLEMENTATION OF ARTIFICIAL INTELLIGENCE